

1. Which three bones comprise the complex synovial joint of the knee?
 - Answer: Femur, Tibia, and Patella.
2. True or False: The knee can support the body in an erect position without muscular activity.
 - Answer: True.
3. What factor compromises the stability of the knee joint?
 - Answer: The long lever arms of the femur and the tibia.
4. Which bony feature is found on the distal femur?
 - Answer: Medial and Lateral Epicondyles.
5. The "Apex" refers to which part of the patella?
 - Answer: The inferior pole.
6. Which of the following is NOT a palpable feature of the proximal tibia or fibula?
 - Answer: Medial Epicondyle (this is on the femur).
7. What is the primary mechanical function of the patella regarding the knee extensors?
 - Answer: To improve efficiency and increase torque by increasing the moment arm.
8. Which muscle's forces are centralized by the patella into one concerted direction of pull?
 - Answer: Quadriceps.
9. How does the patella protect the femoral condyles?
 - Answer: By providing bony protection from direct trauma when the knee is flexed.
10. The knee complex consists of which two primary joints?
 - Answer: Patellofemoral and Tibiofemoral.
11. Why are knee flexion and extension not considered "pure" rolling or hinge motions?
 - Answer: Because the femoral condyles are convex both ways, articulating with tibial condyles that have only slight concavity.
12. What structure separates the femoral condyles posteriorly and serves as the site for cruciate ligaments?
 - Answer: Intercondylar fossa.
13. Which ligament connects the anterior horns of the two menisci?
 - Answer: Transverse ligament.
14. Which muscle sends fibers to the posterior edge of the medial meniscus?
 - Answer: Semimembranosus.
15. The Medial Meniscus is described as being what shape?
 - Answer: C-shaped.
16. The Lateral Meniscus is described as being what shape?
 - Answer: O-shaped.
17. Which ligament anchors the menisci to the lateral rim of the tibia?
 - Answer: Coronary ligament.
18. How much movement occurs in the lateral meniscus compared to the medial?
 - Answer: 12 mm (lateral) vs 6 mm (medial).
19. If the menisci are removed, by how much does the magnitude of stress on the load-bearing surface increase?
 - Answer: Three times.
20. Which structure prevents the joint capsule from intruding into the joint space?
 - Answer: The menisci.
21. Which group of ligaments provides

- anterior-posterior stability?
- Answer: **Cruciate ligaments.**
22. **True or False: Both the collateral and cruciate ligaments are taut in full extension.**
- Answer: **True.**
23. **The Medial Collateral Ligament (MCL) protects the knee against which type of force?**
- Answer: **Valgus force.**
24. **The Lateral Collateral Ligament (LCL) protects the knee against which type of force?**
- Answer: **Varus force.**
25. **The LCL attaches to which bony structure?**
- Answer: **The head of the fibula.**
26. **Which ligament prevents anterior displacement of the tibia on the femur?**
- Answer: **Anterior Cruciate Ligament (ACL).**
27. **The ACL protects the knee against which excessive movement?**
- Answer: **Hyperextension.**
28. **Which ligament prevents posterior displacement of the tibia on the femur?**
- Answer: **Posterior Cruciate Ligament (PCL).**
29. **The PCL protects the knee against which excessive movement?**
- Answer: **Hyperflexion.**
30. **What is the function of the Ligament of Humphrey?**
- Answer: **It anchors the lateral meniscus.**
31. **What is the normal range for knee flexion?**
- Answer: **125-150 degrees.**
32. **What is the normal range for knee hyperextension?**
- Answer: **15 degrees.**
33. **When does the femur rotate on a fixed tibia (Closed Chain axial rotation)?**
- Answer: **When turning from kneeling, sitting, or squatting positions.**
34. **During Terminal Knee Extension in OPEN CHAIN, what movement does the tibia perform?**
- Answer: **Lateral rotation (about 20 degrees).**
35. **During knee flexion in CLOSED CHAIN, how does the femur move?**
- Answer: **It glides anteriorly and rotates externally.**
36. **What is the "Close packed position" of the tibiofemoral joint?**
- Answer: **Full extension.**
37. **What is the "Open packed position" of the tibiofemoral joint?**
- Answer: **25 degrees of flexion.**
38. **The patella is what type of bone?**
- Answer: **Sesamoid bone.**
39. **Which specific muscle is the dynamic stabilizer of the patella from 20 to 0 degrees?**
- Answer: **VMO (Vastus Medialis Obliquus).**
40. **As the knee moves from extension into flexion, the contact area on the patella moves in which direction?**
- Answer: **From the inferior surface to the superior surface.**
41. **What is the "Close packed position" of the patellofemoral joint?**
- Answer: **Full flexion.**
42. **What is the "Open pack position" of the patellofemoral joint?**
- Answer: **Full extension.**
43. **How is the Q Angle measured?**
- Answer: **Line from ASIS to center of patella, and a line**

**from patella to tibial
tuberosity.**

- 44. What is the normal Q Angle range for women?**
- Answer: **15 - 23 degrees.**
- 45. Which muscle is considered the "workhorse" of the knee extensors?**
- Answer: **Vastus Intermedius.**
- 46. Besides knee flexion, the hamstrings help stabilize the knee by restricting what?**
- Answer: **Anterior glide of the tibia on the femur.**
- 47. Which muscle acts to "unlock" the knee during the initiation of flexion?**
- Answer: **Popliteus.**
- 48. What is "Active Insufficiency" of the rectus femoris?**
- Answer: **It prevents full combined hip flexion and knee extension.**
- 49. In CLOSED CHAIN, what occurs to allow the body to rise from a sitting position?**
- Answer: **Cocontraction of the hamstrings and quadriceps.**
- 50. What is "Extensor Lag"?**
- Answer: **Inability to achieve full active knee extension despite having full passive motion.**